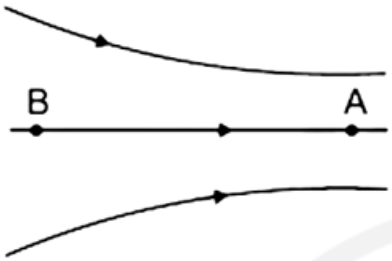
Lakshya JEE 2.0 (2025)

Physics

DPP: 7

Electric Charges and Fields

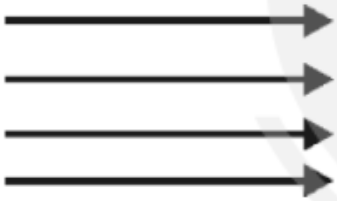
Q1 Some electric lines of force are shown in figure, for point A and B.



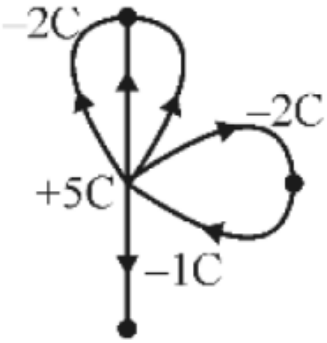
(A) $E_A > E_B$
(B) $E_A < E_B$
(C) $E_A = E_B$
(D) Cant determine from the given information

Q2 Which of the following configurations of electric lines of force is not possible?

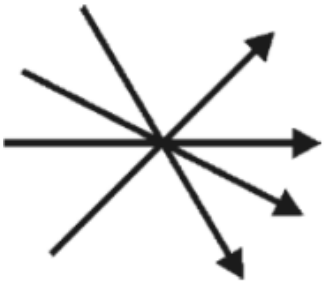
(A)



(B)

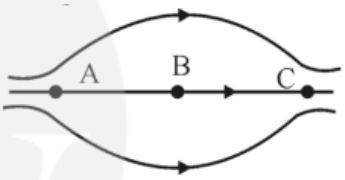


(C)



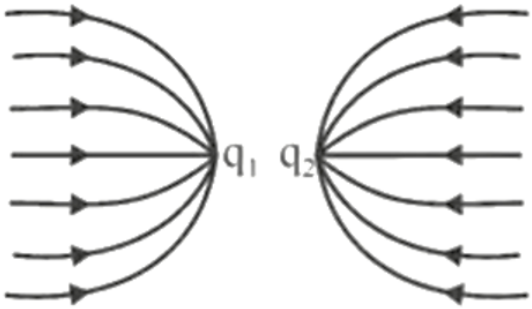
(D) Both (B) and (C)

Q3 The figure shows some of the electric field lines corresponding to an electric field. The figure suggests



(A) $E_{\mathrm{A}} > E_B > E_C$
(B) $E_A = E_B = E_C$
(C) $E_A = E_C > E_B$
(D) $E_A = E_C < E_B$

Q4 The given figure gives electric lines of force due to two charges q_1 and q_2. What are the signs of the two charges?

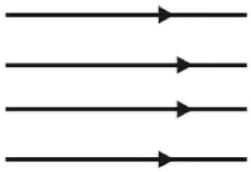


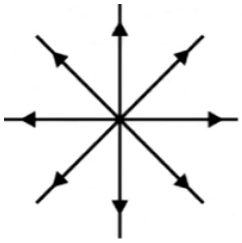
(A) Both are negative

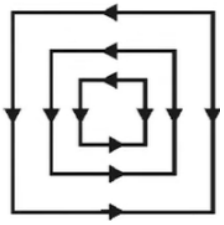
(B) Both are positive
(C) q_1 is positive but q_2 is negative
(D) q_1 is negative but q_2 is positive

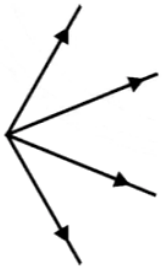
Q5 Choose the correct statements:
(A) The density of electric lines of force at a point is independent of the magnitude of electric intensity vector E at the point
(B) The density of electric lines of force at a point is proportional to the magnitude of electric intensity vector E at that point
(C) Actually, the electric field lines do not exist. This is just a graphical description of the electric field
(D) Actually, the electric field lines exist

Q6 Which one of the followed pattern of electric lines of force cannot be possible for the field produced by a static charge-

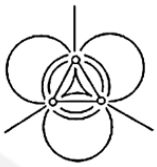
(A)


(B)


(C)


(D)


Q7 Three positive charge of equal value q are placed at the vertices of an equilateral triangle. The resulting lines of force should sketched as in:

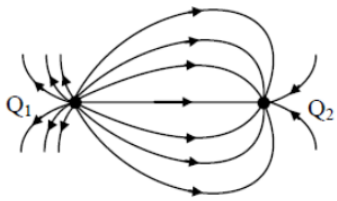
(A)


(B)

(C)

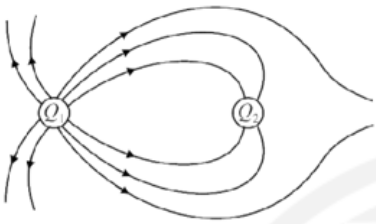
(D)

Q8 A few electric field lines for a system of two charges Q_1 and Q_2 fixed at two different points on the x-axis are shown in the figure. These lines suggest that :

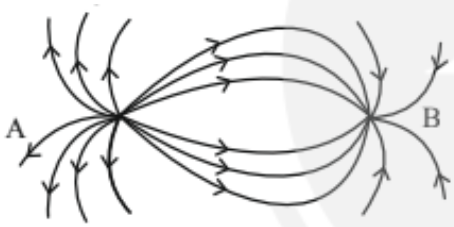


(A) $|Q_1| > |Q_2|$
(B) $|Q_1| < |Q_2|$
(C) at a finite distance to the left of Q_1 the electric field is zero
(D) at a finite distance to the right of Q_2 the electric field is zero

Q9 Find the value of $\frac{Q_1}{Q_2}$ by looking at the various electric lines of forces between charges Q_1 and Q_2.



Q10 The spatial distribution of the electric field lines due to charges (A, B) is shown in figure. Which one of the following statements is correct?



(A) A is +ve and B is -ve and $|A| > |B|$
(B) A is -ve and B is +ve; $|A| = |B|$
(C) Both are +ve but $\mathrm{A} > \mathrm{B}$
(D) Both are -ve A $>$ B

Answer Key

Q1	(A)	Q6	(C)
Q2	(D)	Q7	(C)
Q3	(C)	Q8	(A, D)
Q4	(A)	Q9	-2.5
Q5	(B, C)	Q10	(A)

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